Group 16

October 5th, 2025

Blackstone Inc.

Time Series Economics & Forecasting

PRESENTED BY

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Introduction



Asset Series Chosen: Blackstone Inc.

The dataset chosen is a time series of Blackstone Inc.'s stock prices. Blackstone is one of the world's leading alternative asset management firms, specializing in private equity, real estate, hedge fund solutions, and credit. Headquartered in New York, the firm manages investments across global markets and is known for its strategic focus on long-term value creation and portfolio diversification. The dataset includes the past stock prices of the asset. The objective of this report is to analyze key trends in the stock's performance, compute log returns, conduct the statistical analysis, and forecast future stock prices using the ARIMA model.

Start and End Date: 19th September 2024 – 18th September 2022

Interpolation & Data Cleaning

HistoricalData_175826

Date	Price
09/18/2025	$188.68
09/17/2025	$183.62
09/16/2025	$183.72
09/15/2025	$183.80
09/12/2025	$181.35
09/11/2025	$183.92
09/10/2025	$178.23
09/09/2025	$173.17
09/08/2025	$173.30
09/05/2025	$169.99
09/04/2025	$169.90
09/03/2025	$167.34
09/02/2025	$169.55
08/29/2025	$171.40

Our actual price series does contain all 250 values. This screenshot is provided as reference.

The dataset used for the analysis consists of 250 daily observations of Blackstone Asset Management's stock prices. The data has been pre-adjusted to include only trading days, hence weekend and non-trading days have been removed. A thorough check was performed for missing or null values in the stock's prices.

The results confirmed that the dataset is complete with no missing observations, and therefore no interpolation or imputation techniques were required. The dataset is thus ready for further analysis such as return generation, visualization, and stationarity testing without any additional cleaning procedures.

Interpolation & Data Cleaning (Cont'd)

Series: BX_NEW Workfile: BX_NEW::Bx_new\

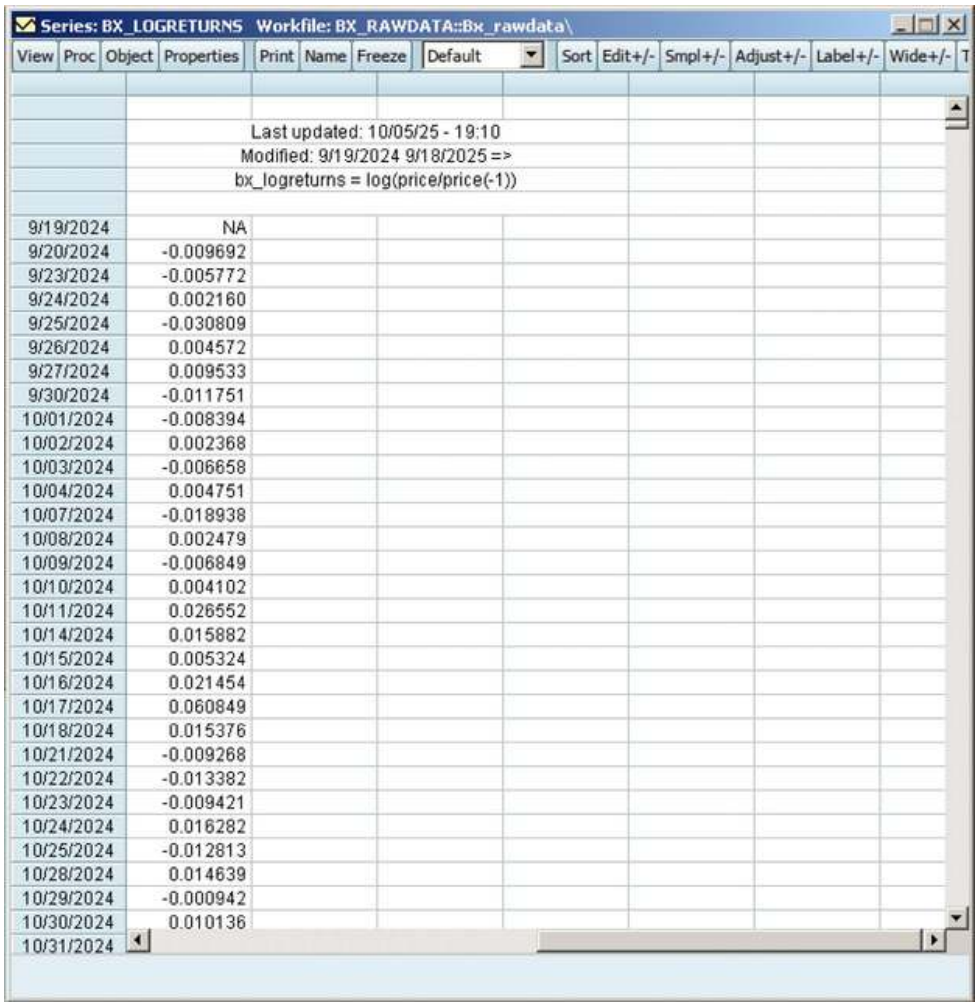
View | Proc | Object | Properties | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+

BX_New

8/14/2025	174.68
8/15/2025	171.62
8/18/2025	169.22
8/19/2025	167.29
8/20/2025	164.47
8/21/2025	163.44
8/22/2025	170.50
8/25/2025	169.50
8/26/2025	171.43
8/27/2025	171.64
8/28/2025	172.02
8/29/2025	171.40
9/02/2025	169.55
9/03/2025	167.34
9/04/2025	169.90
9/05/2025	169.99
9/08/2025	173.30
9/09/2025	173.17
9/10/2025	178.23
9/11/2025	183.92
9/12/2025	181.35
9/15/2025	183.80
9/16/2025	183.72
9/17/2025	183.62
9/18/2025	188.68

Now we have directly labelled the raw data as the new data using the formula:-
genr BX_New=price

Generating Log Returns

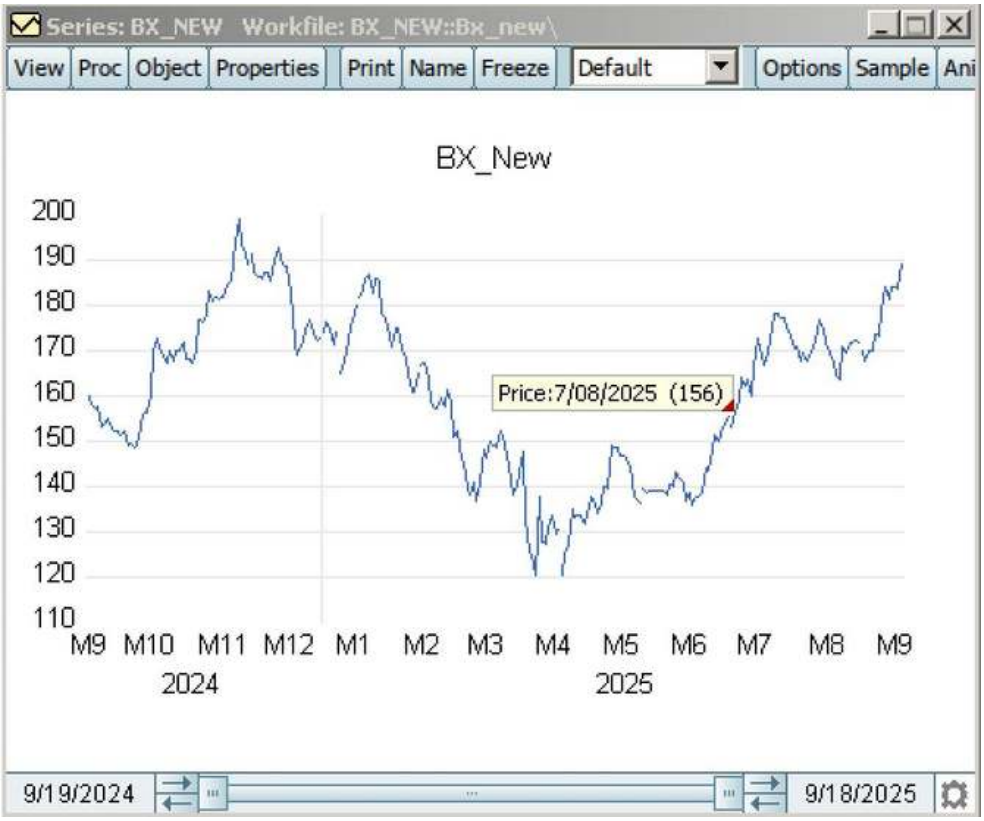


9/19/2024	NA
9/20/2024	-0.009692
9/23/2024	-0.005772
9/24/2024	0.002160
9/25/2024	-0.030809
9/26/2024	0.004572
9/27/2024	0.009533
9/30/2024	-0.011751
10/01/2024	-0.008394
10/02/2024	0.002368
10/03/2024	-0.006658
10/04/2024	0.004751
10/07/2024	-0.018938
10/08/2024	0.002479
10/09/2024	-0.006849
10/10/2024	0.004102
10/11/2024	0.026552
10/14/2024	0.015882
10/15/2024	0.005324
10/16/2024	0.021454
10/17/2024	0.060849
10/18/2024	0.015376
10/21/2024	-0.009268
10/22/2024	-0.013382
10/23/2024	-0.009421
10/24/2024	0.016282
10/25/2024	-0.012813
10/28/2024	0.014639
10/29/2024	-0.000942
10/30/2024	0.010136
10/31/2024	

Now, we have generated the log returns of the series. Log returns are used in time series because of the following reasons: (i)interpreted as continuously compounded returns (ii)Mathematical Ease (iii)Log Normality (iv)Time Additivity (v)Approximate Raw-Log Equality.

The formulas that we have used to generate the log returns are :- genr BX_LogReturns= log(price/price(-1))

Data Visualization of Price Series

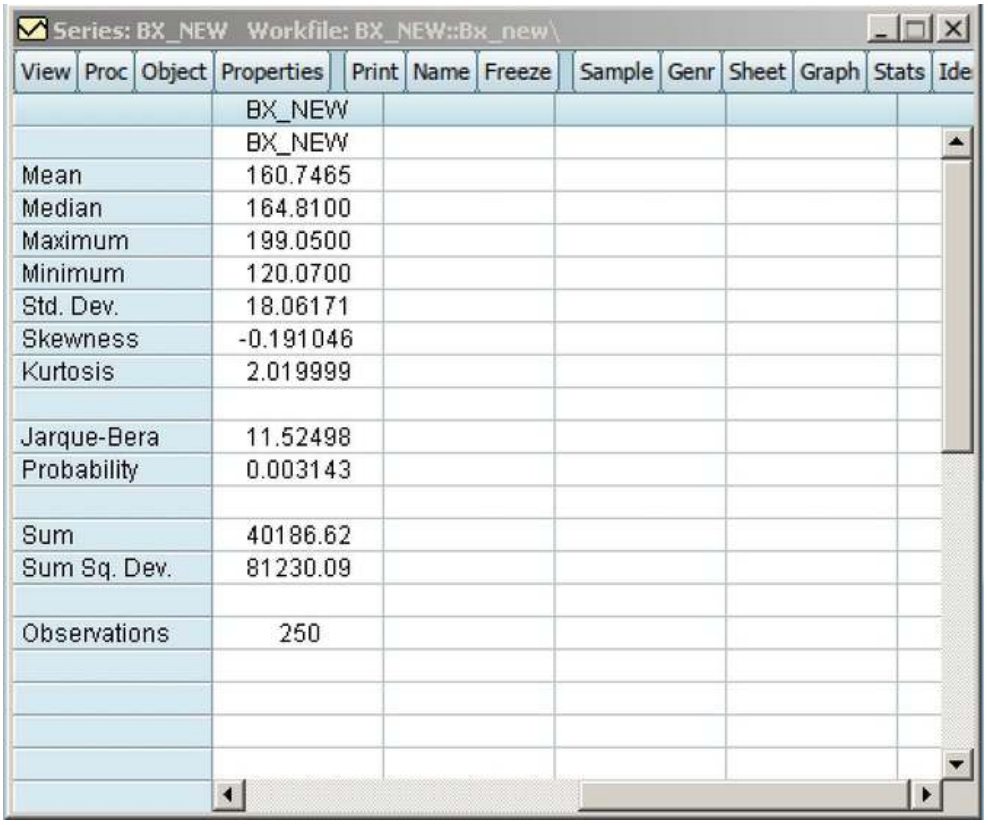


This graph shows the daily adjusted closing prices of Blackstone Asset Management from September 2024 to September 2025.

Observations and trends:

- The price starts around ₹150–160 in late 2024, rises toward ₹190, and then declines sharply to around ₹120 by March 2025.
- After March 2025, the series begins to recover steadily, showing an upward trend until September 2025, reaching nearly ₹190–200 again.
- The pattern indicates cyclical movement with phases of rise and fall — consistent with market corrections and recoveries.
- Overall, the series is non-stationary, as both the mean and variance change over time (long-term upward drift and fluctuating volatility).

Descriptive Statistics (Price Series)



	BX_NEW
	BX_NEW
Mean	160.7465
Median	164.8100
Maximum	199.0500
Minimum	120.0700
Std. Dev.	18.06171
Skewness	-0.191046
Kurtosis	2.019999
Jarque-Bera	11.52498
Probability	0.003143
Sum	40186.62
Sum Sq. Dev.	81230.09
Observations	250

Mean: The average value of the series is approximately 160.7465.

Median: The median price is 164.8100; since the mean is less than the median, this indicates a slightly negatively-skewed distribution.

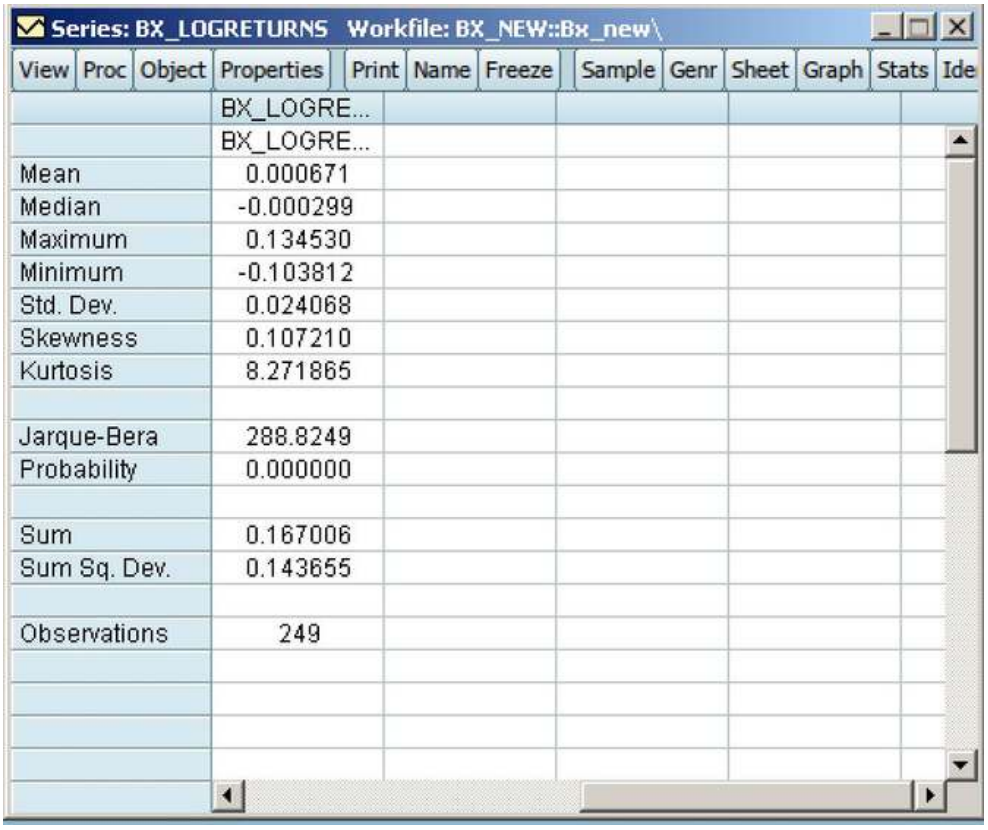
Standard Deviation: A standard deviation of $18.06171 indicates the degree of price variability or volatility over the period.

Skewness: The skewness value of -0.191046 confirms a slight negative skew, though its proximity to zero suggests the distribution is nearly symmetrical.

Kurtosis: The kurtosis value of $2.019999 (less than 3) suggests a platykurtic distribution, which is flatter and has thinner tails than a normal distribution.

Jarque-Bera Test: The test statistic of $11.52498 and a very low probability (0.003143) strongly suggest that the data does not follow a normal distribution.

Descriptive Statistics (Log Returns)



	BX_LOGRE...
	BX_LOGRE...
Mean	0.000671
Median	-0.000299
Maximum	0.134530
Minimum	-0.103812
Std. Dev.	0.024068
Skewness	0.107210
Kurtosis	8.271865
Jarque-Bera	288.8249
Probability	0.000000
Sum	0.167006
Sum Sq. Dev.	0.143655
Observations	249

Mean: The average log return is 0.000671, indicating a small but positive return on average over the period.

Standard Deviation: The standard deviation of $0.024068 measures the volatility or risk associated with the returns.

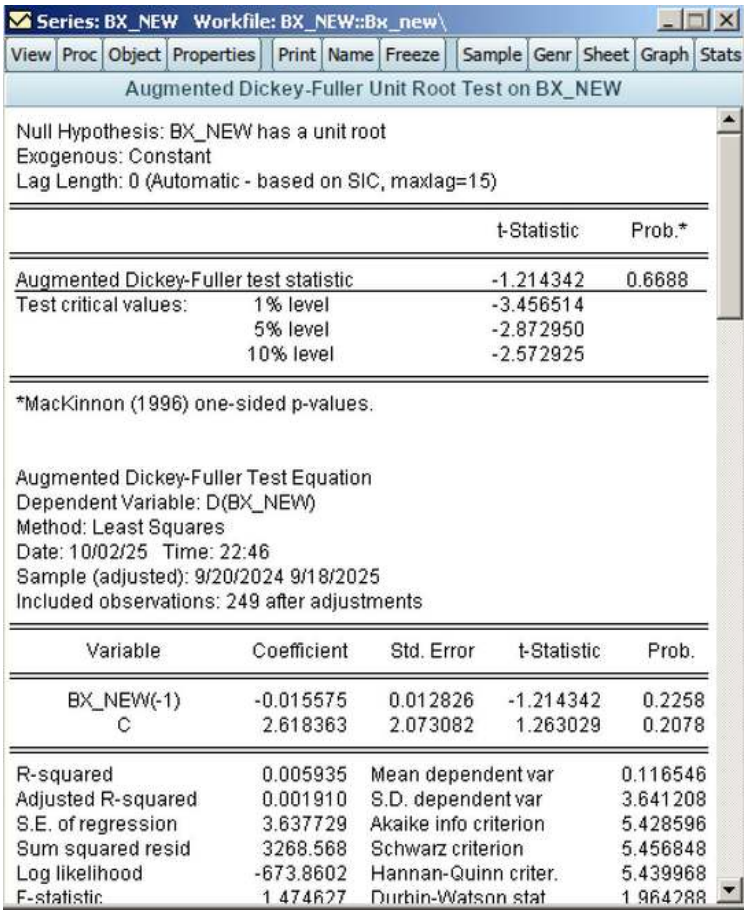
Skewness: A skewness of $0.107210 indicates that the distribution of returns is slightly positively skewed.

Kurtosis: The very high kurtosis value of $8.271865 (much greater than 3) indicates a leptokurtic distribution with "fat tails," meaning extreme price movements occur more frequently than a normal distribution would predict.

Jarque-Bera Test: An extremely high test statistic (288.8249) and a probability of $0.000000 provide overwhelming evidence that the returns series is not normally distributed.

Stationary Tests for Price Series

ADF Test (At Level)



Series: BX_NEW Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats

Augmented Dickey-Fuller Unit Root Test on BX_NEW

Null Hypothesis: BX_NEW has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=15)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.214342	0.6688
Test critical values:	1% level	-3.456514	
	5% level	-2.872950	
	10% level	-2.572925	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(BX_NEW)
Method: Least Squares
Date: 10/02/25 Time: 22:46
Sample (adjusted): 9/20/2024 9/18/2025
Included observations: 249 after adjustments

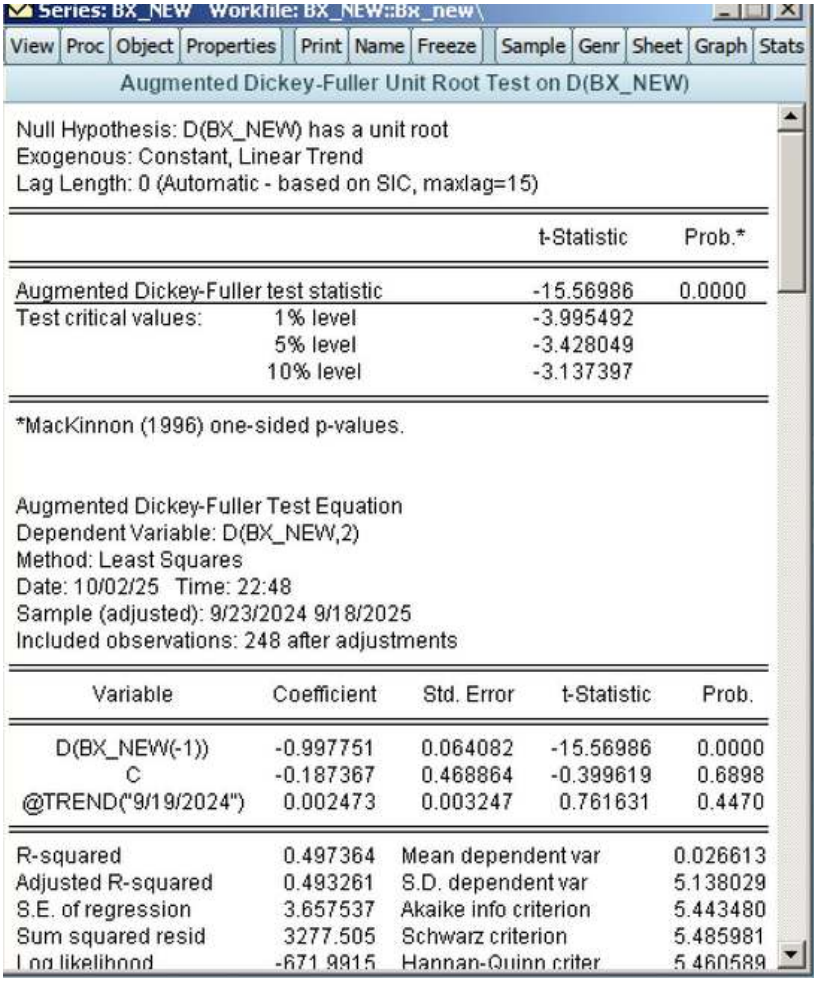
Variable	Coefficient	Std. Error	t-Statistic	Prob.
BX_NEW(-1)	-0.015575	0.012826	-1.214342	0.2258
C	2.618363	2.073082	1.263029	0.2078

R-squared	0.005935	Mean dependent var	0.116546
Adjusted R-squared	0.001910	S.D. dependent var	3.641208
S.E. of regression	3.637729	Akaike info criterion	5.428596
Sum squared resid	3268.568	Schwarz criterion	5.456848
Log likelihood	-673.8602	Hannan-Quinn criter.	5.439968
F-statistic	1.474627	Durbin-Watson stat	1.964288

Test statistic < Critical value - condition
-1.214342 > 1% (-3.456514), 5% (-2.872950), and 10% (-2.572925)
→ Fail to reject null
P-value < Chosen significance level
0.6688 > 0.01, 0.05, and 0.10
→ Fail to reject null

Analysis: The Augmented Dickey-Fuller (ADF) test statistic of -1.214342 is greater than the critical values at all conventional significance levels. The p-value (0.6688) is also much higher than 0.05, confirming that the null hypothesis of a unit root cannot be rejected. The series is non-stationary.

ADF Test (At First Difference)



Series: BX_NEW Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats

Augmented Dickey-Fuller Unit Root Test on D(BX_NEW)

Null Hypothesis: D(BX_NEW) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=15)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-15.56986	0.0000
Test critical values:	1% level	-3.995492	
	5% level	-3.428049	
	10% level	-3.137397	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(BX_NEW,2)
Method: Least Squares
Date: 10/02/25 Time: 22:48
Sample (adjusted): 9/23/2024 9/18/2025
Included observations: 248 after adjustments

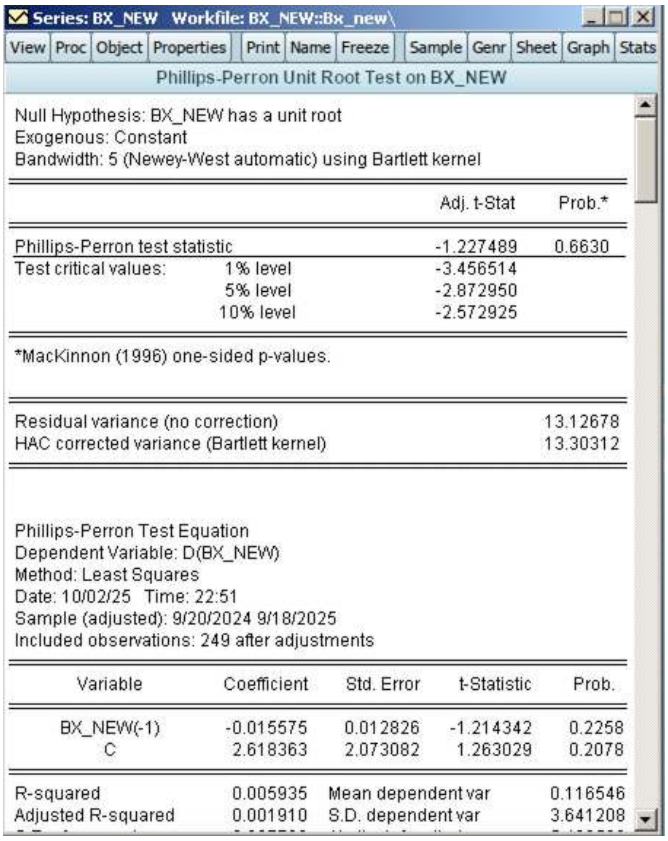
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(BX_NEW(-1))	-0.997751	0.064082	-15.56986	0.0000
C	-0.187367	0.468864	-0.399619	0.6898
@TREND("9/19/2024")	0.002473	0.003247	0.761631	0.4470

R-squared	0.497364	Mean dependent var	0.026613
Adjusted R-squared	0.493261	S.D. dependent var	5.138029
S.E. of regression	3.657537	Akaike info criterion	5.443480
Sum squared resid	3277.505	Schwarz criterion	5.485981
Log likelihood	-671.9915	Hannan-Quinn criter	5.460589

Test statistic < Critical value - condition
-15.56986 < 1% (-3.995492), 5% (-3.428049), and 10% (-3.137397)
→ Reject null
P-value < Chosen significance level
0.0000 < 0.01, 0.05, and 0.10
→ Reject null

Analysis:
At the first difference, the test statistic is far less than all the critical values, and the p-value is effectively zero. This allows rejection of the null hypothesis of a unit root. The series is now stationary.

PP (At Level)



Series: BX_NEW Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats

Phillips-Perron Unit Root Test on BX_NEW

Null Hypothesis: BX_NEW has a unit root
Exogenous: Constant
Bandwidth: 5 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-1.227489	0.6630
Test critical values:	1% level	-3.456514	
	5% level	-2.872950	
	10% level	-2.572925	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	13.12678
HAC corrected variance (Bartlett kernel)	13.30312

Phillips-Perron Test Equation
Dependent Variable: D(BX_NEW)
Method: Least Squares
Date: 10/02/25 Time: 22:51
Sample (adjusted): 9/20/2024 9/18/2025
Included observations: 249 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
BX_NEW(-1)	-0.015575	0.012826	-1.214342	0.2258
C	2.618363	2.073082	1.263029	0.2078

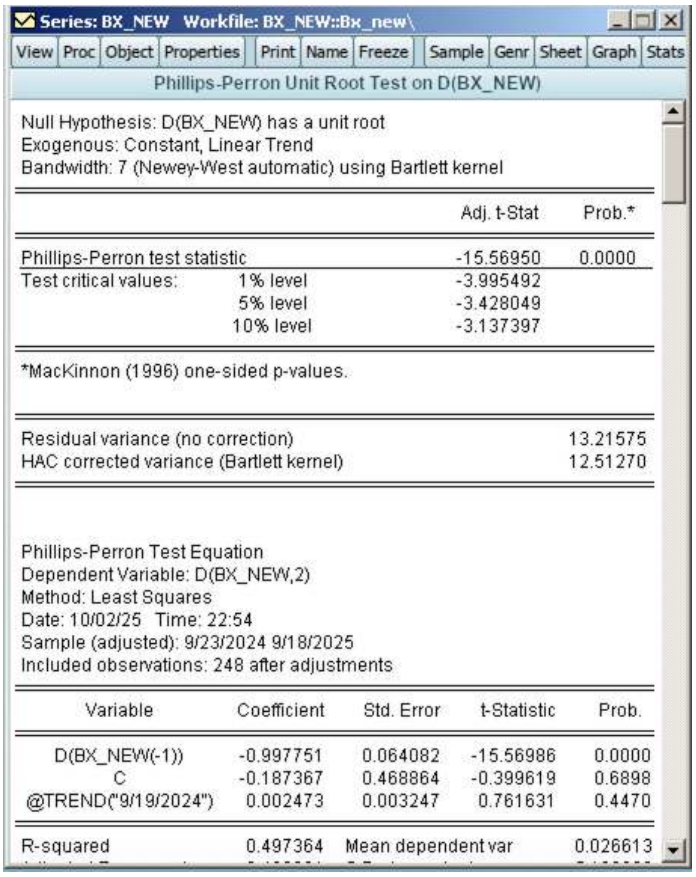
R-squared	0.005935	Mean dependent var	0.116546
Adjusted R-squared	0.001910	S.D. dependent var	3.641208

Test Statistic: -1.227489
Critical Values:

1%: -3.456514
5%: -2.872950
10%: -2.572925
Condition: -1.227489 > -3.456514, -2.872950, -2.572925 → Fail to Reject Null
p-value: 0.6630 > 0.10 → Fail to Reject Null

Analysis: At level with the intercept, the null hypothesis of a unit root cannot be rejected. The series BX_NEW is non-stationary at level.

PP (At First Difference)



Series: BX_NEW Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats

Phillips-Perron Unit Root Test on D(BX_NEW)

Null Hypothesis: D(BX_NEW) has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 7 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-15.56950	0.0000
Test critical values:	1% level	-3.995492	
	5% level	-3.428049	
	10% level	-3.137397	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	13.21575
HAC corrected variance (Bartlett kernel)	12.51270

Phillips-Perron Test Equation
Dependent Variable: D(BX_NEW,2)
Method: Least Squares
Date: 10/02/25 Time: 22:54
Sample (adjusted): 9/23/2024 9/18/2025
Included observations: 248 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(BX_NEW(-1))	-0.997751	0.064082	-15.56986	0.0000
C	-0.187367	0.468864	-0.399619	0.6898
@TREND("9/19/2024")	0.002473	0.003247	0.761631	0.4470

R-squared	0.497364	Mean dependent var	0.026613

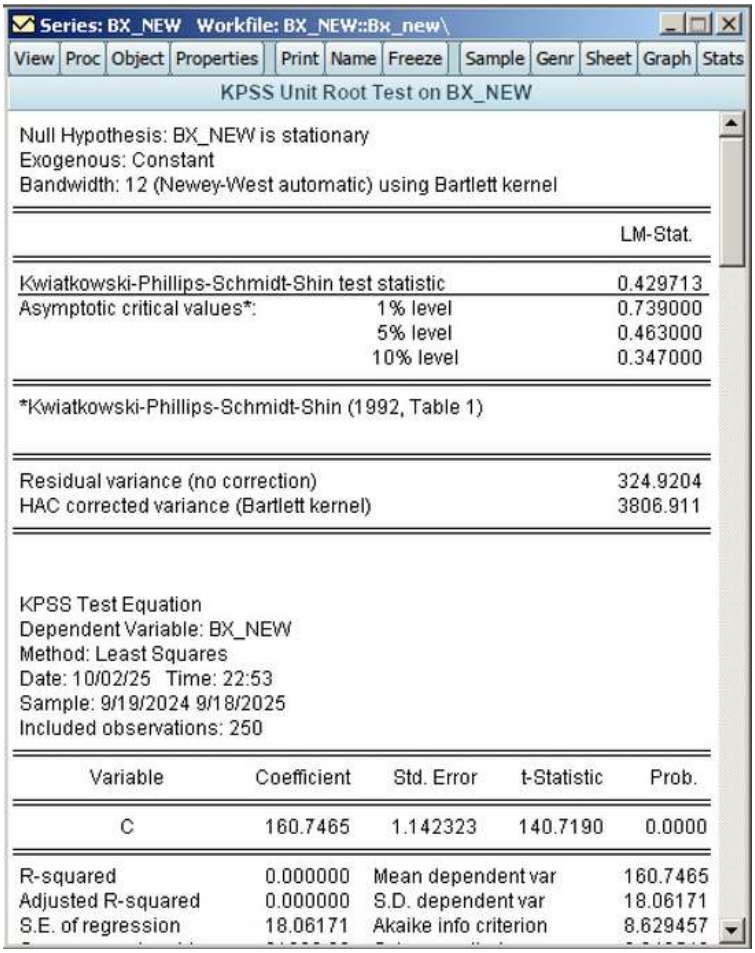
Phillips–Perron Unit Root Test (First Difference – Intercept & Trend)

Test Statistic: -15.56950
Critical Values:

1%: -3.995492
5%: -3.428049
10%: -3.137397
Condition: -15.56950 < -3.995492, -3.428049, -3.137397 → Reject Null
p-value: 0.0000 < 0.01 → Reject Null

Analysis: At first difference with intercept and trend, the null hypothesis of a unit root is rejected. The series D(BX_NEW) is stationary.

KPSS (At Level)



Series: BX_NEW Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats

KPSS Unit Root Test on BX_NEW

Null Hypothesis: BX_NEW is stationary
Exogenous: Constant
Bandwidth: 12 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.429713
Asymptotic critical values*:	1% level	0.739000
	5% level	0.463000
	10% level	0.347000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	324.9204
HAC corrected variance (Bartlett kernel)	3806.911

KPSS Test Equation
Dependent Variable: BX_NEW
Method: Least Squares
Date: 10/02/25 Time: 22:53
Sample: 9/19/2024 9/18/2025
Included observations: 250

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	160.7465	1.142323	140.7190	0.0000

R-squared	0.000000	Mean dependent var	160.7465
Adjusted R-squared	0.000000	S.D. dependent var	18.06171
S.E. of regression	18.06171	Akaike info criterion	8.629457

KPSS Test at Level (Intercept)
Test statistic: 0.429713
Critical values:
1%: 0.739000
5%: 0.463000
10%: 0.347000

0.429713 < 1% (0.739000) → Fail to reject null
0.429713 < 5% (0.463000) → Fail to reject null
0.429713 > 10% (0.347000) → Reject null at 10% level

Analysis: At the 1% and 5% levels, the test statistic is less than the critical values, meaning we fail to reject the null hypothesis that BX_NEW is stationary. However, at the 10% level, we reject the null.

KPSS (At First Difference)

Series: BX_NEW Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats

KPSS Unit Root Test on D(BX_NEW)

Null Hypothesis: D(BX_NEW) is stationary
Exogenous: Constant, Linear Trend
Bandwidth: 7 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.130701
Asymptotic critical values*:	1% level	0.216000
	5% level	0.146000
	10% level	0.119000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	13.16999
HAC corrected variance (Bartlett kernel)	12.54916

KPSS Test Equation
Dependent Variable: D(BX_NEW)
Method: Least Squares
Date: 10/02/25 Time: 22:55
Sample (adjusted): 9/20/2024 9/18/2025
Included observations: 249 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.209521	0.463216	-0.452317	0.6514
@TREND("9/19/2024")	0.002609	0.003212	0.812003	0.4176

R-squared	0.002662	Mean dependent var	0.116546
Adjusted R-squared	-0.001375	S.D. dependent var	3.641208

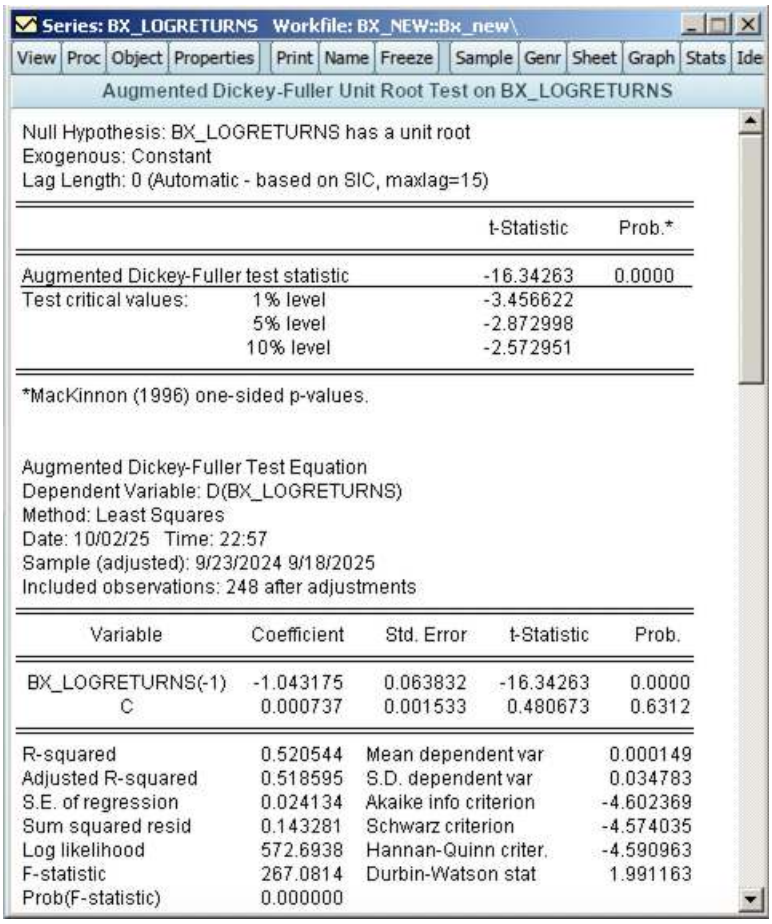
Test statistic: 0.130701
Critical values:
1%: 0.216000
5%: 0.146000
10%: 0.119000

Comparison:
0.130701 < 1% (0.216000) → Fail to reject null
0.130701 < 5% (0.146000) → Fail to reject null
0.130701 > 10% (0.119000) → Reject null at 10% level

Analysis: The test statistic is below the 1% and 5% critical values, indicating we fail to reject the null hypothesis of stationarity at those levels. It's only slightly above the 10% critical value.

Stationary Tests for Log Returns

ADF (At Level)



Series: BX_LOGRETURNS Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ide

Augmented Dickey-Fuller Unit Root Test on BX_LOGRETURNS

Null Hypothesis: BX_LOGRETURNS has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=15)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-16.34263	0.0000
Test critical values:	1% level	-3.456622	
	5% level	-2.872998	
	10% level	-2.572951	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(BX_LOGRETURNS)
Method: Least Squares
Date: 10/02/25 Time: 22:57
Sample (adjusted): 9/23/2024 9/18/2025
Included observations: 248 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
BX_LOGRETURNS(-1)	-1.043175	0.063832	-16.34263	0.0000
C	0.000737	0.001533	0.480673	0.6312

R-squared	0.520544	Mean dependent var	0.000149
Adjusted R-squared	0.518595	S.D. dependent var	0.034783
S.E. of regression	0.024134	Akaike info criterion	-4.602369
Sum squared resid	0.143281	Schwarz criterion	-4.574035
Log likelihood	572.6938	Hannan-Quinn criter.	-4.590963
F-statistic	267.0814	Durbin-Watson stat	1.991163
Prob(F-statistic)	0.000000		

Level with Intercept

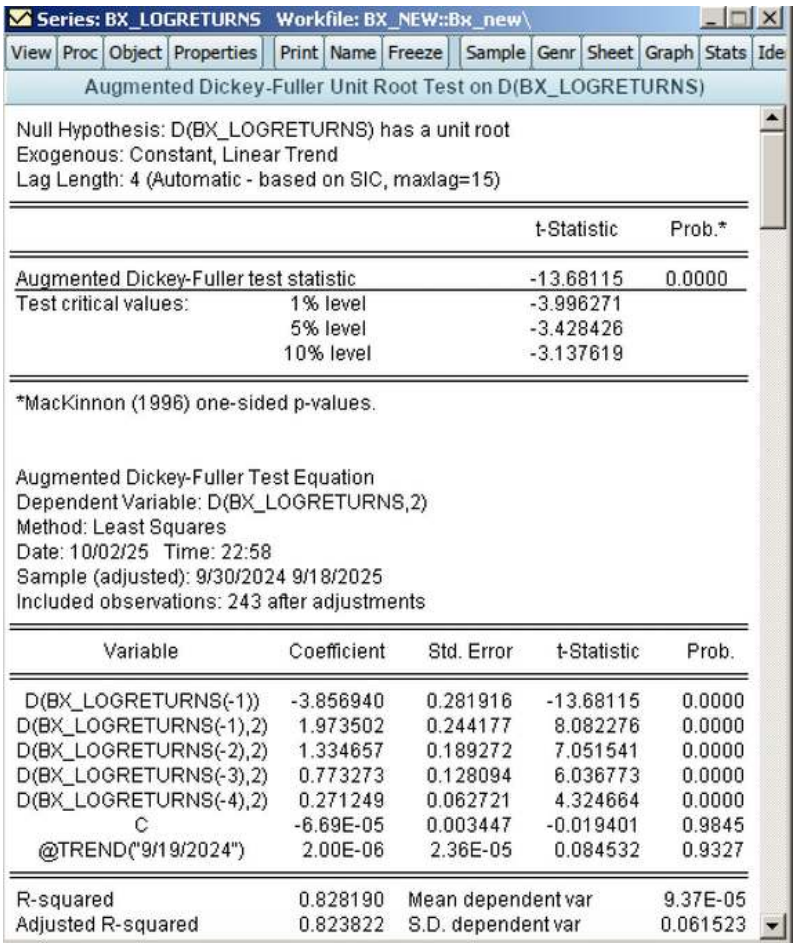
Test statistic = -16.34263

Critical values = -3.456622 (1%), -2.872998 (5%), -2.572951 (10%)

P-value = 0.0000

Since the test statistic (-16.34263) is less than the critical values at 1%, 5%, and 10% levels, and the p-value is less than 0.01, we reject the null hypothesis. This means the series BX_LOGRETURNS is stationary at the level with intercept.

ADF (At First Difference)



Series: BX_LOGRETURNS Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ide

Augmented Dickey-Fuller Unit Root Test on D(BX_LOGRETURNS)

Null Hypothesis: D(BX_LOGRETURNS) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 4 (Automatic - based on SIC, maxlag=15)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-13.68115	0.0000
Test critical values:	1% level	-3.996271	
	5% level	-3.428426	
	10% level	-3.137619	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(BX_LOGRETURNS,2)
Method: Least Squares
Date: 10/02/25 Time: 22:58
Sample (adjusted): 9/30/2024 9/18/2025
Included observations: 243 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(BX_LOGRETURNS(-1))	-3.856940	0.281916	-13.68115	0.0000
D(BX_LOGRETURNS(-1),2)	1.973502	0.244177	8.082276	0.0000
D(BX_LOGRETURNS(-2),2)	1.334657	0.189272	7.051541	0.0000
D(BX_LOGRETURNS(-3),2)	0.773273	0.128094	6.036773	0.0000
D(BX_LOGRETURNS(-4),2)	0.271249	0.062721	4.324664	0.0000
C	-6.69E-05	0.003447	-0.019401	0.9845
@TREND("9/19/2024")	2.00E-06	2.36E-05	0.084532	0.9327

R-squared	0.828190	Mean dependent var	9.37E-05
Adjusted R-squared	0.823822	S.D. dependent var	0.061523

First Difference with Intercept and trend

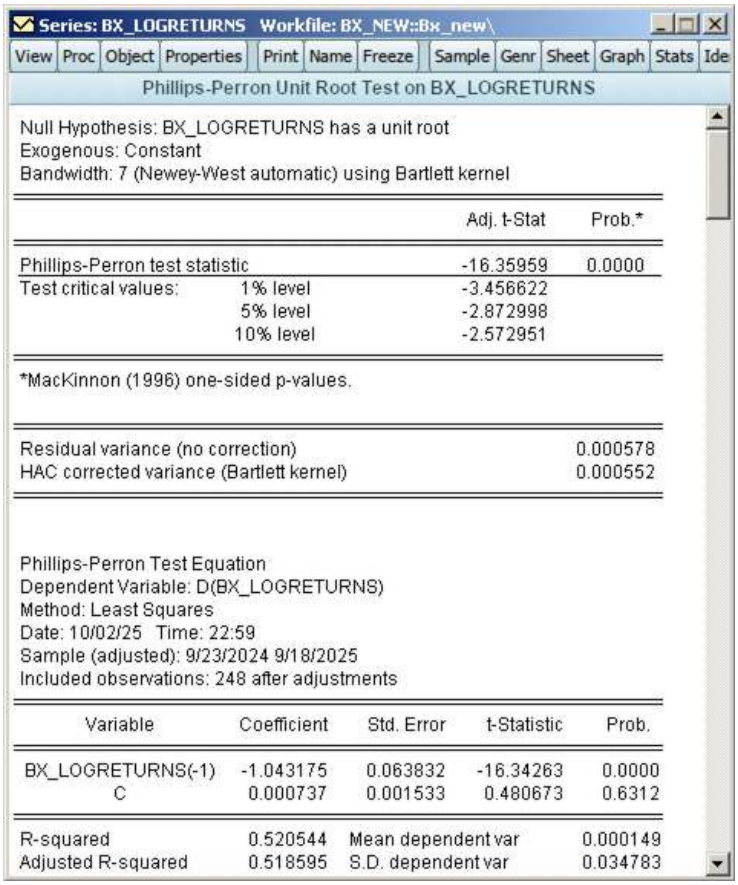
Test statistic = -13.68115

Critical values = -3.996271 (1%), -3.428426 (5%), -3.137619 (10%)

P-value = 0.0000

Since the test statistic (-13.68115) is less than all critical values and the p-value is less than 0.01, we reject the null hypothesis. This means the first differenced series is stationary with intercept.

PP (At Level)



Series: BX_LOGRETURNS Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ide

Phillips-Perron Unit Root Test on BX_LOGRETURNS

Null Hypothesis: BX_LOGRETURNS has a unit root
Exogenous: Constant
Bandwidth: 7 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-16.35959	0.0000
Test critical values:	1% level	-3.456622	
	5% level	-2.872998	
	10% level	-2.572951	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.000578
HAC corrected variance (Bartlett kernel)	0.000552

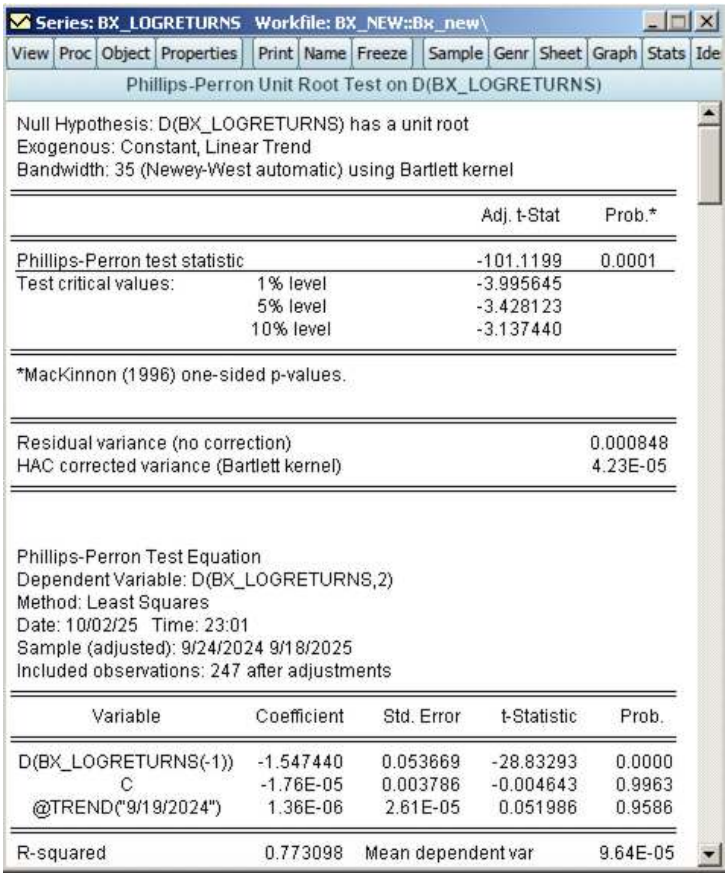
Phillips-Perron Test Equation
Dependent Variable: D(BX_LOGRETURNS)
Method: Least Squares
Date: 10/02/25 Time: 22:59
Sample (adjusted): 9/23/2024 9/18/2025
Included observations: 248 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
BX_LOGRETURNS(-1)	-1.043175	0.063832	-16.34263	0.0000
C	0.000737	0.001533	0.480673	0.6312

R-squared	0.520544	Mean dependent var	0.000149
Adjusted R-squared	0.518595	S.D. dependent var	0.034783

Test statistic: −16.35959
Critical values: 1% (−3.456622-), 5% (−2.872998), 10% (−2.572951)
P-value: 0.0000
Analysis:
Test statistic −16.35959 < critical values at 1%, 5%, and 10% significance levels (−3.456622, −2.872998, −2.572951) → Reject the null hypothesis of a unit root.
P-value 0.0000< chosen significance levels → Reject the null hypothesis.

PP (At First Difference)



Series: BX_LOGRETURNS Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ide

Phillips-Perron Unit Root Test on D(BX_LOGRETURNS)

Null Hypothesis: D(BX_LOGRETURNS) has a unit root
Exogenous: Constant, Linear Trend
Bandwidth: 35 (Newey-West automatic) using Bartlett kernel

		Adj. t-Stat	Prob.*
Phillips-Perron test statistic		-101.1199	0.0001
Test critical values:	1% level	-3.995645	
	5% level	-3.428123	
	10% level	-3.137440	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.000848
HAC corrected variance (Bartlett kernel)	4.23E-05

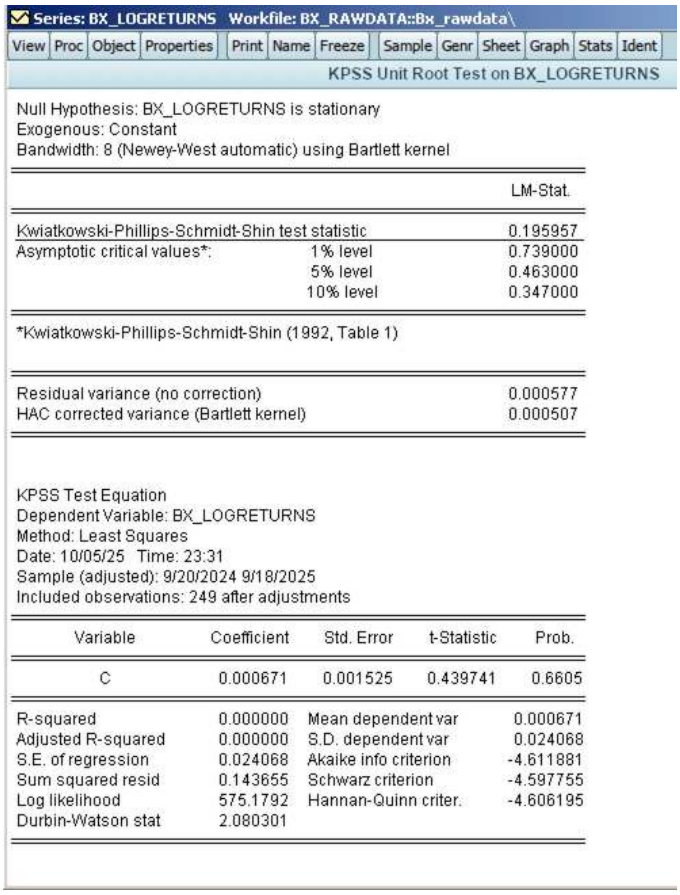
Phillips-Perron Test Equation
Dependent Variable: D(BX_LOGRETURNS,2)
Method: Least Squares
Date: 10/02/25 Time: 23:01
Sample (adjusted): 9/24/2024 9/18/2025
Included observations: 247 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(BX_LOGRETURNS(-1))	-1.547440	0.053669	-28.83293	0.0000
C	-1.76E-05	0.003786	-0.004643	0.9963
@TREND("9/19/2024")	1.36E-06	2.61E-05	0.051986	0.9586
R-squared	0.773098	Mean dependent var		9.64E-05

Test statistic: −101.1199
Critical values: 1% (−3.995645), 5% (−3.428123), 10% (−3.137440
P-value: 0.0001

Analysis:
Test statistic −101.1199 < critical values at 1%, 5%, and 10% significance levels (−3.995645, −3.428123, −3.137440) → Reject the null hypothesis of a unit root.
P-value 0.0001 < chosen significance levels → Reject the null hypothesis.

KPSS (At Level)



Series: BX_LOGRETURNS Workfile: BX_RAWDATA::Bx_rawdata\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

KPSS Unit Root Test on BX_LOGRETURNS

Null Hypothesis: BX_LOGRETURNS is stationary
Exogenous: Constant
Bandwidth: 8 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.195957
Asymptotic critical values*:	1% level	0.739000
	5% level	0.463000
	10% level	0.347000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	0.000577
HAC corrected variance (Bartlett kernel)	0.000507

KPSS Test Equation
Dependent Variable: BX_LOGRETURNS
Method: Least Squares
Date: 10/05/25 Time: 23:31
Sample (adjusted): 9/20/2024 9/18/2025
Included observations: 249 after adjustments

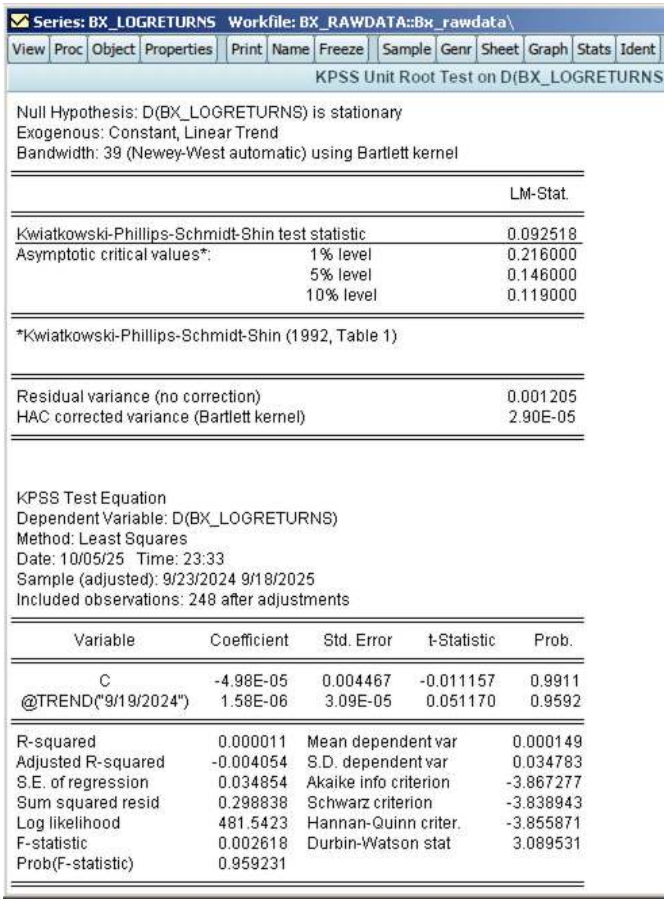
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000671	0.001525	0.439741	0.6605

R-squared	0.000000	Mean dependent var	0.000671
Adjusted R-squared	0.000000	S.D. dependent var	0.024068
S.E. of regression	0.024068	Akaike info criterion	-4.611881
Sum squared resid	0.143655	Schwarz criterion	-4.597755
Log likelihood	575.1792	Hannan-Quinn criter.	-4.606195
Durbin-Watson stat	2.080301		

KPSS Test at Level (Intercept)
Test statistic (LM-Stat): 0.195957
Critical values: 1% (0.739000), 5% (0.463000), 10% (0.347000)

Analysis: Test statistic 0.1959570.1959570.195957 < critical values at 1%, 5%, and 10% significance levels (0.739000, 0.463000, 0.347000) → Fail to reject the null hypothesis of stationarity. The results indicate that the BX_LOGRETURNS series is stationary at level with intercept according to the KPSS test.

KPSS (At First Difference)



Series: BX_LOGRETURNS Workfile: BX_RAWDATA::Bx_rawdata\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

KPSS Unit Root Test on D(BX_LOGRETURNS)

Null Hypothesis: D(BX_LOGRETURNS) is stationary
Exogenous: Constant, Linear Trend
Bandwidth: 39 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.092518
Asymptotic critical values*:	1% level	0.216000
	5% level	0.146000
	10% level	0.119000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	0.001205
HAC corrected variance (Bartlett kernel)	2.90E-05

KPSS Test Equation
Dependent Variable: D(BX_LOGRETURNS)
Method: Least Squares
Date: 10/05/25 Time: 23:33
Sample (adjusted): 9/23/2024 9/18/2025
Included observations: 248 after adjustments

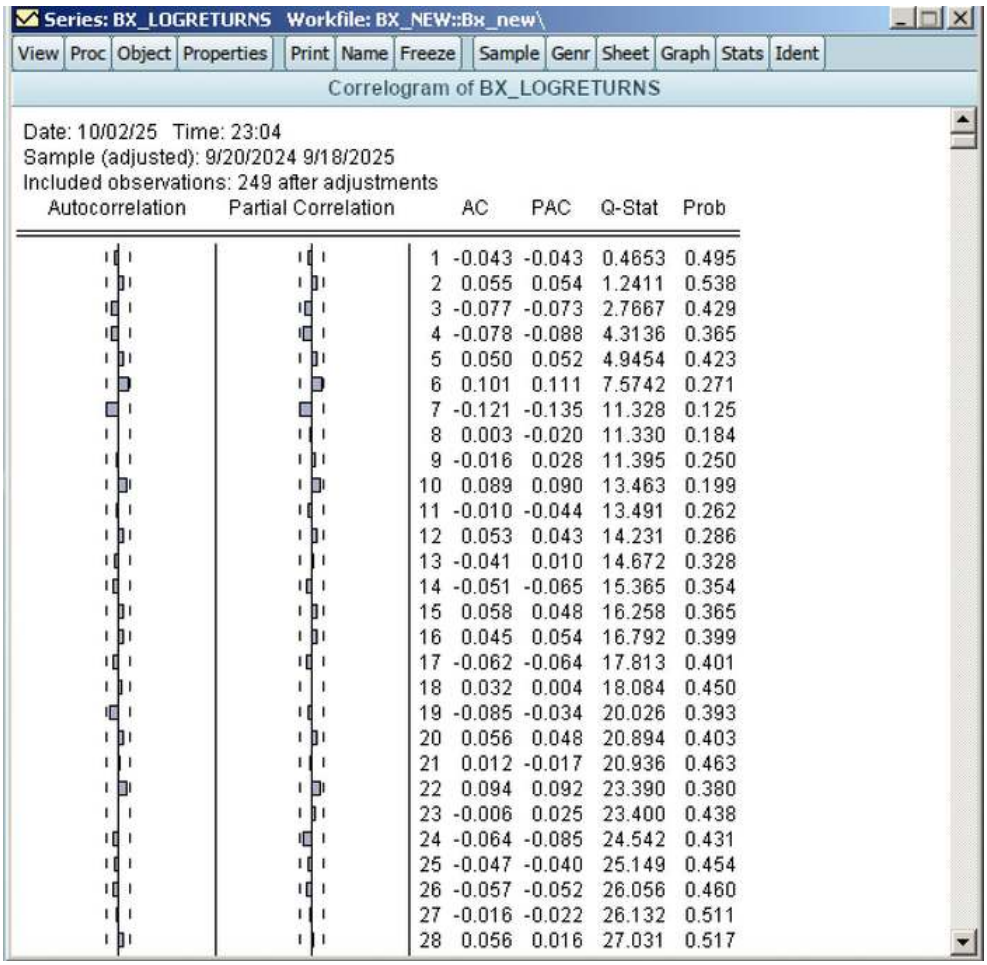
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-4.98E-05	0.004467	-0.011157	0.9911
@TREND("9/19/2024")	1.58E-06	3.09E-05	0.051170	0.9592

R-squared	0.000011	Mean dependent var	0.000149
Adjusted R-squared	-0.004054	S.D. dependent var	0.034783
S.E. of regression	0.034854	Akaike info criterion	-3.867277
Sum squared resid	0.298838	Schwarz criterion	-3.838943
Log likelihood	481.5423	Hannan-Quinn criter.	-3.855871
F-statistic	0.002618	Durbin-Watson stat	3.089531
Prob(F-statistic)	0.959231		

Test statistic (LM-Stat): 0.092518
Critical values: 1% (0.216000), 5% (0.146000), 10% (0.119000)

Analysis: Test statistic 0.0925180.0925180.092518 < critical values at 1%, 5%, and 10% significance levels (0.216000, 0.146000, 0.119000) → Fail to reject the null hypothesis of stationarity. The results indicate that the first differenced BX_LOGRETURNS series is stationary when including intercept and trend.

ACF and PACF



	AC	PAC	Q-Stat	Prob
1	-0.043	-0.043	0.4653	0.495
2	0.055	0.054	1.2411	0.538
3	-0.077	-0.073	2.7667	0.429
4	-0.078	-0.088	4.3136	0.365
5	0.050	0.052	4.9454	0.423
6	0.101	0.111	7.5742	0.271
7	-0.121	-0.135	11.328	0.125
8	0.003	-0.020	11.330	0.184
9	-0.016	0.028	11.395	0.250
10	0.089	0.090	13.463	0.199
11	-0.010	-0.044	13.491	0.262
12	0.053	0.043	14.231	0.286
13	-0.041	0.010	14.672	0.328
14	-0.051	-0.065	15.365	0.354
15	0.058	0.048	16.258	0.365
16	0.045	0.054	16.792	0.399
17	-0.062	-0.064	17.813	0.401
18	0.032	0.004	18.084	0.450
19	-0.085	-0.034	20.026	0.393
20	0.056	0.048	20.894	0.403
21	0.012	-0.017	20.936	0.463
22	0.094	0.092	23.390	0.380
23	-0.006	0.025	23.400	0.438
24	-0.064	-0.085	24.542	0.431
25	-0.047	-0.040	25.149	0.454
26	-0.057	-0.052	26.056	0.460
27	-0.016	-0.022	26.132	0.511
28	0.056	0.016	27.031	0.517

What the correlogram says:
From the ACF/PACF of BX_LOGRETURNS

The PACF has clearly visible spikes at lags 6 and 7, just outside the 95% band. Most other partial autocorrelations are small and inside the band.
The ACF is small everywhere (largest |AC| around lags 6–7) and decays quickly with mild oscillation, no long tail, and no sharp cut-off at low lags.

Interpretation: this is the classic ARMA fingerprint, neither a pure AR (PACF clean cut-off with long ACF tail) nor a pure MA (ACF cut-off with long PACF tail). The isolated signal around lags 6–7 suggests short memory with a weak “weekly/7-step” effect. That motivates models that include dynamics up to about 6–7 lags on at least one side (AR or MA)

The Chosen Model

Equation: UNTITLED Workfile: BX_NEW::Bx_new\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: BX_LOGRETURNS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 10/02/25 Time: 23:33
Sample: 9/20/2024 9/18/2025
Included observations: 249
Convergence achieved after 15 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000660	0.001449	0.455490	0.6492
AR(6)	0.094226	0.067346	1.399134	0.1630
MA(7)	-0.131293	0.052396	-2.505762	0.0129
SIGMASQ	0.000562	2.86E-05	19.62878	0.0000

R-squared	0.025830	Mean dependent var	0.000671
Adjusted R-squared	0.013902	S.D. dependent var	0.024068
S.E. of regression	0.023900	Akaike info criterion	-4.613251
Sum squared resid	0.139945	Schwarz criterion	-4.556746
Log likelihood	578.3497	Hannan-Quinn criter.	-4.590507
F-statistic	2.165395	Durbin-Watson stat	2.068024
Prob(F-statistic)	0.092684		

Inverted AR Roots	.67	.34-.58i	.34+.58i	-.34-.58i
	-.34+.58i	-.67		
Inverted MA Roots	.75	.47+.58i	.47-.58i	-.17+.73i
	-.17-.73i	-.67-.32i	-.67+.32i	

Forecasting

Equation: UNTITLED Workfile: BX_NEW::Bx_new\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: BX_LOGRETURNS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 10/02/25 Time: 23:33
Sample: 9/20/2024 9/18/2025
Included observations: 249
Convergence achieved after 28 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000644	0.001285	0.501426	0.6165
AR(7)	0.186288	0.421606	0.441853	0.6590
MA(7)	-0.317905	0.406031	-0.782958	0.4344
SIGMASQ	0.000566	2.67E-05	21.25180	0.0000

R-squared	0.018146	Mean dependent var	0.000671
Adjusted R-squared	0.006123	S.D. dependent var	0.024068
S.E. of regression	0.023994	Akaike info criterion	-4.605542
Sum squared resid	0.141049	Schwarz criterion	-4.549036
Log likelihood	577.3899	Hannan-Quinn criter.	-4.582797
F-statistic	1.509281	Durbin-Watson stat	2.050455
Prob(F-statistic)	0.212680		

Inverted AR Roots	.79	.49+.61i	.49-.61i	-.18-.77i
	-.18+.77i	-.71-.34i	-.71+.34i	
Inverted MA Roots	.85	.53+.66i	.53-.66i	-.19+.83i
	-.19-.83i	-.76-.37i	-.76+.37i	

Equation: UNTITLED Workfile: BX_NEW::Bx_new\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: BX_LOGRETURNS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 10/02/25 Time: 23:32
Sample: 9/20/2024 9/18/2025
Included observations: 249
Convergence achieved after 14 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000661	0.001473	0.448769	0.6540
AR(7)	-0.115714	0.052093	-2.221274	0.0272
MA(6)	0.086930	0.066696	1.303379	0.1937
SIGMASQ	0.000564	2.85E-05	19.79812	0.0000

R-squared	0.023122	Mean dependent var	0.000671
Adjusted R-squared	0.011160	S.D. dependent var	0.024068
S.E. of regression	0.023933	Akaike info criterion	-4.610617
Sum squared resid	0.140334	Schwarz criterion	-4.554112
Log likelihood	578.0218	Hannan-Quinn criter.	-4.587873
F-statistic	1.933007	Durbin-Watson stat	2.069349
Prob(F-statistic)	0.124800		

Inverted AR Roots	.66+.32i	.66-.32i	.16+.72i	.16-.72i
	-.46-.57i	-.46+.57i	-.73	
Inverted MA Roots	.58-.33i	.58+.33i	.00-.67i	.00+.67i
	-.58+.33i	-.58-.33i		

Equation: UNTITLED Workfile: BX_NEW::Bx_new\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: BX_LOGRETURNS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 10/02/25 Time: 23:28
Sample: 9/20/2024 9/18/2025
Included observations: 249
Convergence achieved after 20 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000688	0.001771	0.388312	0.6981
AR(6)	0.417556	0.503345	0.829562	0.4076
MA(6)	-0.318278	0.523412	-0.608082	0.5437
SIGMASQ	0.000570	3.11E-05	18.34362	0.0000

R-squared	0.012193	Mean dependent var	0.000671
Adjusted R-squared	0.000098	S.D. dependent var	0.024068
S.E. of regression	0.024067	Akaike info criterion	-4.599735
Sum squared resid	0.141904	Schwarz criterion	-4.543230
Log likelihood	576.6671	Hannan-Quinn criter.	-4.576991
F-statistic	1.008091	Durbin-Watson stat	2.062205
Prob(F-statistic)	0.389798		

Inverted AR Roots	.86	.43+.75i	.43-.75i	-.43+.75i
	-.43-.75i	-.86		
Inverted MA Roots	.83	.41-.72i	.41+.72i	-.41+.72i
	-.41-.72i	-.83		

Equation: UNTITLED Workfile: BX_NEW::Bx_new\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: BX_LOGRETURNS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 10/02/25 Time: 23:33
Sample: 9/20/2024 9/18/2025
Included observations: 249
Convergence achieved after 15 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000660	0.001449	0.455490	0.6492
AR(6)	0.094226	0.067346	1.399134	0.1630
MA(7)	-0.131293	0.052396	-2.505762	0.0129
SIGMASQ	0.000562	2.86E-05	19.62878	0.0000

R-squared	0.025830	Mean dependent var	0.000671
Adjusted R-squared	0.013902	S.D. dependent var	0.024068
S.E. of regression	0.023900	Akaike info criterion	-4.613251
Sum squared resid	0.139945	Schwarz criterion	-4.556746
Log likelihood	578.3497	Hannan-Quinn criter.	-4.590507
F-statistic	2.165395	Durbin-Watson stat	2.068024
Prob(F-statistic)	0.092684		

Inverted AR Roots	.67	.34-.58i	.34+.58i	-.34-.58i
	-.34+.58i	-.67		
Inverted MA Roots	.75	.47+.58i	.47-.58i	-.17+.73i
	-.17-.73i	-.67-.32i	-.67+.32i	

Forecasting (Cont'd)

The most crucial step in forecasting a time series is the selection of an appropriate model that accurately represents the underlying data pattern. In this case, since the series is identified as non-stationary, an ARIMA (AutoRegressive Integrated Moving Average) model is considered suitable. The ARIMA framework is particularly effective for financial time series like stock prices because it captures both the autocorrelation structure of the data and the dependence of current values on past observations. By incorporating lagged terms of the series (autoregressive part), differencing to remove non-stationarity (integrated part), and lagged forecast errors (moving average part), the model can generate reliable and dynamic forecasts. Based on the correlograms and unit root test results conducted prior, applying a first-order differencing was found to be sufficient to achieve stationarity. This transformation ensures that the mean and variance of the series remain constant over time, allowing the ARIMA model to be effectively fitted and used for accurate prediction of future stock price movements.

<u>The four candidates and what each misses</u>.

ARMA(6,6): stops at 6 on both sides, so it risks missing the PACF spike at lag 7 entirely.

ARMA(7,6): captures AR(7) but omits MA(7), which is often needed when the ACF shows a small oscillatory pattern beyond 6.

ARMA(7,7): captures everything but adds an extra parameter relative to ARMA(6,7).

ARMA(6,7): includes the MA(7) term to absorb the 7-step feature and keeps AR order at 6, which is consistent with the generally small PACF at lags ≤ 6.

In this step, the 4 potential models will be compared and chosen based on the following criterion:
Lowest AIC, Schwarz Criterion and HQ is better.
Low volatility – low $\sigma 2$
Highest adjusted R2

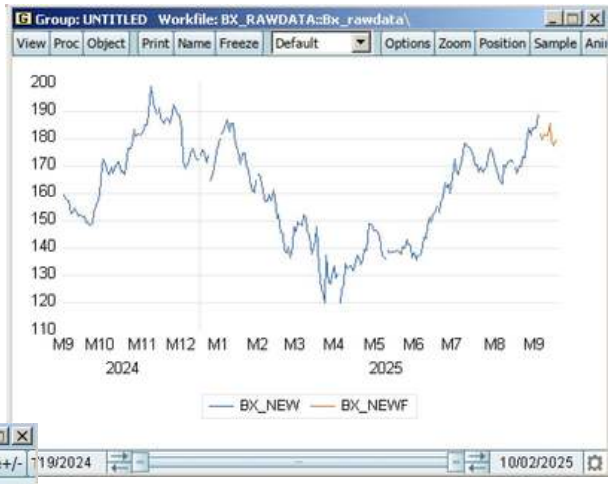
Forecasting (Cont'd)

Series: PRICE Workfile: BX_NEW::Bx_new\

View | Proc | Object | Properties | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Ad

Price

	Price
8/29/2025	171.40
9/02/2025	169.55
9/03/2025	167.34
9/04/2025	169.90
9/05/2025	169.99
9/08/2025	173.30
9/09/2025	173.17
9/10/2025	178.23
9/11/2025	183.92
9/12/2025	181.35
9/15/2025	183.80
9/16/2025	183.72
9/17/2025	183.62
9/18/2025	188.68
9/19/2025	NA
9/22/2025	NA
9/23/2025	NA
9/24/2025	NA
9/25/2025	NA
9/26/2025	NA
9/29/2025	NA
9/30/2025	NA
10/01/2025	NA
10/02/2025	NA



Group: UNTITLED Workfile: BX_RAWDATA::Bx_rawdata\

View | Proc | Object | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Compare+/- | 19/2024 | 10/02/2025

	BX_NEW	BX_NEWF
9/09/2025	173.17	NA
9/10/2025	178.23	NA
9/11/2025	183.92	NA
9/12/2025	181.35	NA
9/15/2025	183.80	NA
9/16/2025	183.72	NA
9/17/2025	183.62	NA
9/18/2025	188.68	NA
9/19/2025	NA	181.6026
9/22/2025	NA	179.2772
9/23/2025	NA	181.4588
9/24/2025	NA	181.3410
9/25/2025	NA	181.2536
9/26/2025	NA	185.6900
9/29/2025	NA	179.3037
9/30/2025	NA	177.2687
10/01/2025	NA	179.1779
10/02/2025	NA	179.0748

The ARIMA(6,1,7) model has passed the diagnostic tests and is ready to forecast. The stock prices of the next 10 days .i.e., from 19th September 20245to 2 October 2025 assuming 6 business days a week have been forecasted using this model. In order to forecast, the following steps have been followed on EViews:

- The next 10 days have been added to the current dataset using Structure/Resize Current page option
- Then, the values have been forecasted with the AR(6) MA(7) structure

We have chosen only 6 and 7 lags, because of the law of parsimony. There are other significant lags at other levels, but we ignore them to keep the model less complex.